

Anqi Xu

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Senior Computer Vision & Robotics Engineer with 15+ years turning perception, learning, and control algorithms into production systems on real hardware. End-to-end builder taking ideas from prototype to scale: Ph.D. research in interactive robot vision, applied ML solutions shipped at a startup, and Meta-scale deployment across partner teams and factories.

WORK EXPERIENCE

Meta (Reality Labs)

02/2021 - Present

Senior Computer Vision Engineer

Redmond, WA

- Production vision systems: Helped move a product launch metric from **57% below to 42% above the market leader** by building a hardware-in-the-loop system that combined robot arms, **50-Gbps multi-camera ingestion**, visual localization, and world-locked rendering to quantify AR/MR geometric calibration and rendering performance.

Tech: C++, Python, OpenCV, OpenXR, sub-ms sensor/display synchronization, robot-arm trajectory design.

- Factory-scale metrology: Deployed calibration and validation systems across **6 AR/MR programs**, cutting error by **up to 70%** and run-to-run variance **10x** using privileged calibration priors; held accuracy across device variations in factory operation via GR&R acceptance limits, thermal-drift modeling, and root-cause analysis.

Tech: Camera/IMU/display calibration, Ceres, GR&R, thermal modeling, telemetry, factory automation.

- Reliability engineering: Accelerated a critical-path system from **140 to 17 seconds without accuracy loss** via parallelism and pipelining, added end-to-end runtime regression tests, and led a dependency-safe monorepo modernization that removed **2M+ lines of dead code** and raised team test coverage from **23% to 68%**.

Tech: CI/CD, multithreading, performance profiling, hermetic builds, unit and regression testing.

- Modular ingestion platform: Eliminated an extra PC hop in an AR/MR data-capture framework by upgrading legacy transport with native Android modules, simplifying the runtime architecture, deployment, and debugging.

Tech: gRPC/protobuf, publisher/subscriber, Android, Linux, Windows.

- Agentic engineering: Compressed a factory build from **6 weeks to 2.5 weeks** using spec-driven development, automated release packaging, and deterministic validation; built reusable agent skills adopted across **20 engineers and 3 partner teams**.

Tech: Context/memory management, multi-agent orchestration, agent skills development, HIL debugging.

- Technical leadership: Owned architecture and roadmap across **2 production workstreams over 5 years**; led 4 engineers through releases aligned with feature, device, and factory partners; mentored interns, contributing to **2 full-time conversions**.

Element AI

03/2017 - 12/2020

Applied Research Scientist, Robotics Lead

Montreal, Canada

- Deep reinforcement learning: Led an applied ML team with McKinsey QuantumBlack, built scaled infrastructure for training and tuning **PPO→SAC** control policies using a high-fidelity sailing simulator, then extended to RL-assisted vehicle co-design; learned sailing strategies contributed to Emirates Team New Zealand's **2021 America's Cup victory**.

Tech: Ray/Tune/RLlib, PyTorch, Docker, Python, AWS SageMaker/S3, large-scale experiment orchestration.

- Sim-to-real robustness: Published ([ICCV 2019](#)) a method for generating inconspicuous adversarial images purely in simulation; verified transfer by diverting visual trackers off target in real-world monitoring and drone-photography setups.

Tech: PyTorch, adversarial ML, domain randomization, Gazebo, ROS, OpenCV, visual servoing.

- 3D visual inspection: Demonstrated an anomaly-localization prototype at a **NeurIPS 2019** industry booth, showcasing few-shot inspection of natural objects by combining RGB-D reconstruction, object-pose regression, and learned visual embeddings.

Tech: PyTorch, Open3D, instance segmentation, contrastive learning.

SELECTED PROJECTS

- [ueye_cam](#): Open-source C++/ROS driver for IDS cameras (**100+ GitHub forks**) enabling synchronized, high-throughput multi-camera capture for robotics/video analytics.
- Overcooked Bot: Built an interactive game-playing agent spanning CNN→ViT detection, appearance/Kalman tracking, homography localization, PID control, and task planning; developed a PyQt tool for model-bootstrapped data labeling. Demos: [detection](#), [tracking](#).
- Aqua robot platform: Refactored a real-time motor-control API from instantaneous joint commands to smooth, parametric, hydrodynamically efficient gaits; integrated IMU/camera/display drivers and ROS services; led pool and field trials.

SKILLS

- **Computer Vision & 3D Perception:** multi-view calibration, bundle adjustment, camera/lens/display modeling, pose estimation, feature and fiducial detection, tracking.
- **Robotics & Sensors:** ROS, Gazebo, RGB/RGB-D cameras, IMUs, GPS; robot arms, UAVs, marine robots; SLAM/VIO, planning, visual servoing, PID control, embedded electronics.
- **Machine Learning:** PyTorch, deep RL (PPO, SAC), imitation learning, simulation-based training, sim-to-real, adversarial ML, CNNs and vision transformers.
- **Metrology & Evaluation:** metrics, telemetry, and limits design, Gage Repeatability & Reproducibility, thermal-drift modeling, CI-to-dashboard regression monitoring.
- **Agentic Engineering:** spec-driven development, agent orchestration, reusable skills, context/memory management.
- **Software & Systems:** C++, Python, Linux/Windows/Android, git, Docker, gRPC, CI/CD.

SELECTED PUBLICATIONS

- A. Xu and G. Dudek. [Maintaining Efficient Collaboration with Trust-Seeking Robots](#). IROS 2016. **Finalist, IROS KROS Best Paper Award on Cognitive Robotics.**
- D. Meger, J. Gamboa, A. Xu, P. Giguère, and G. Dudek. [Learning Legged Swimming Gaits from Experience](#). ICRA 2015. **Finalist, ICRA Best Conference Paper Award.**

EDUCATION

Ph.D. in Computer Science

McGill University, Canada, 2017

Thesis: [Efficient Collaboration with Trust-Seeking Robots](#)

Vanier Canadian Graduate Scholar • CIPPRS Doctoral Dissertation Award (2016)

B.Eng. in Computer Engineering

McGill University, Canada, 2008

Great Distinction • Dean's Honour List

LEADERSHIP & COMMUNITY

- Program co-chair, [Conference on Computer and Robot Vision \(CRV\)](#), 2018-2019.
- Reviewer for ICRA, IROS, RSS, RA-L, HRI, CoRL, CRV, ICCV, and IJRR.